

```

#include <stdio.h>

#define MAX 5

int queue[MAX];

int front = -1, rear = -1, item, choice;

void insert(int item) {
    if (rear == MAX - 1) {
        printf("Queue is full\n");
    } else {
        if (front == -1) {
            front = 0;
            rear = 0;
            queue[rear] = item;
        } else {
            int j = rear;
            while (j >= front && item < queue[j]) {
                queue[j + 1] = queue[j];
                j--;
            }
            queue[j + 1] = item;
            rear = rear + 1;
        }
        printf("%d is inserted into the queue\n", item);
    }
}

```

```

void delete() {
    if (front == -1 || front > rear) {
        printf("Queue is empty\n");
    } else {
        printf("Deleted item: %d\n", queue[front]);
        if (front == rear) {
            front = -1;
            rear = -1;
        } else {
            front++;
        }
    }
}

void display() {
    if (front == -1 || front > rear) {
        printf("Queue is empty\n");
    } else {
        printf("Queue Elements:\n");
        for (int i = front; i <= rear; i++) {
            printf(" %d", queue[i]);
        }
        printf("\n");
    }
}

int main() {

```

```
while (1) {  
    printf("\nQUEUE OPERATION:\n 1. Insert\n 2. Delete\n 3.  
Display\n 4. Exit\nEnter your choice: ");  
    scanf("%d", &choice);  
    switch (choice) {  
        case 1:  
            printf("Enter value to insert: ");  
            scanf("%d", &item);  
            insert(item);  
            break;  
        case 2:  
            delete();  
            break;  
        case 3:  
            display();  
            break;  
        case 4:  
            printf("Exiting...\n");  
            return 0;  
        default:  
            printf("Invalid choice\n");  
    }  
}  
return 0; }
```

Output

```
QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 12
12 is inserted into the queue

QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 13
13 is inserted into the queue

QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 14
14 is inserted into the queue

QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 15
15 is inserted into the queue

QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 16
16 is inserted into the queue

QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 2
Deleted item: 12

QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 3
Queue Elements:
13 14 15 16

QUEUE OPERATION:
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 4
Exiting...

Process returned 0 (0x0)    execution time : 27.983 s
Press any key to continue.
|
```